CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 25: Complementary Angles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $\dfrac{\cos 37^\circ}{\sin 53^\circ}$.
2. Evaluate $\operatorname{cosec} 31^\circ - \sec 59^\circ$.
3. Evaluate $\dfrac{\tan 26^\circ}{\cot 64^\circ} + \dfrac{\sin 18^\circ}{\cos 72^\circ}$.
4. Evaluate $\cos^2 40^\circ + \cos^2 50^\circ$.
5. Evaluate $\tan 5^\circ \tan 25^\circ \tan 30^\circ \tan 65^\circ \tan 85^\circ$.
6. Express $\cos 78^\circ + \cot 62^\circ$ in terms of angles between 0° and 45°.
7. If $\sin 2A = \cos(A + 15^\circ)$, where $2A$ is acute, find A.
8. If $\tan 3\theta = \cot(\theta + 10^\circ)$, where 3θ is acute, find θ.
9. Prove that $\sin 63^\circ \cos 27^\circ + \cos 63^\circ \sin 27^\circ = 1$.
10. If A, B, C are the angles of a triangle, prove that $\tan \dfrac{A+B}{2} = \cot \dfrac{C}{2}$.

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Answer Key

1. 1
2. 0
3. 2
4. 1
5. $\frac{1}{\sqrt{3}}$
6. $\sin 12^\circ + \tan 28^\circ$
7. 25°
8. 20°
9. It equals $\sin^2 63^\circ + \cos^2 63^\circ = 1$
10. $\frac{A+B}{2} = 90^\circ - \frac{C}{2}$, and $\tan\left(90^\circ - \frac{C}{2}\right) = \cot\frac{C}{2}$

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